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Sheet supply device, recording device, and cart

Abstract

A sheet supply device that supplies a sheet from a roll, includes: a first roll support portion which rotatably supports the roll on one end side in a width direction of the roll; a second roll support portion which rotatably supports the roll on the other end side of the roll by opposing the first roll support portion; and a guide portion which extends in the width direction and movably supports the second roll support portion in the width direction; wherein below a first space in which the roll between the first roll support portion and the second roll support portion is installed, a second space into which a cart which transports the roll enters is formed.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a roll attachment mechanism which attaches a roll to a recording device which performs recording on a sheet while supporting the roll, around which a continuous sheet is wound, by a cart, the recording device, and the cart.

Description of the Related Art

(2) In a recording device which performs a print operation on a sheet supplied from a roll, the roll is set on the device prior to sheet supply. As a device which supports the roll, such a configuration that includes a pair of holders which rotatably holds a paper core of the roll is known. In such configuration, the holder is moved in a width direction of the roll and inserted into the paper core of the roll so that the roll is set on the recording device. Here, when the roll is transported to the recording device prior to the setting of the roll, since particularly a roll with a high weight cannot

be carried directly by hands, the roll is placed on a cart which supports the roll from below for the transportation in many cases as in Japanese Patent No. 6217336.

SUMMARY OF THE INVENTION

(3) The present invention has an object to improve workability of roll attachment.

(4) In order to achieve the object described above, a sheet supply device according to the present invention is a sheet supply device that supplies a sheet from a roll, including:

(5) a first roll support portion which rotatably supports the roll on one end side in a width direction of the roll;

(6) a second roll support portion which is provided to face the first roll support portion and rotatably supports the roll on the other end side of the roll; and

(7) a guide portion which extends in the width direction and movably supports the second roll support portion in the width direction, wherein

(8) below a first space in which the roll between the first roll support portion and the second roll support portion is installed, a second space into which a cart which transports the roll enters is formed.

(9) Moreover, in order to achieve the object described above, the cart according to the present invention is a cart which supports the roll when the roll around which the sheet is wound is to be attached to the recording device, including:

(10) a roll receiving portion which supports the roll from below;

(11) a positioned portion whose position in a crossing direction crossing a width direction of the roll with respect to the recording device is determined by a positioning portion provided in the recording device; and

(12) a position adjustment mechanism capable of adjusting a position in the width direction of the roll supported by the roll receiving portion in a state in which the positioned portion is positioned by the positioning portion.

(13) According to the present invention, workability of roll attachment can be improved.

(14) Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a front view of a recording device;

(2) FIG. **2** is a schematic sectional view of the recording device illustrating a conveyance path of a sheet;

(3) FIG. **3** is a perspective view of a roll set portion according to a first embodiment;

(4) FIG. **4** is a perspective view of a cart according to the first embodiment;

(5) FIGS. **5**A to **5**D are diagrams illustrating a roll setting procedure according to the first embodiment;

(6) FIG. **6** is a schematic sectional view of a recording device and the cart according to the first embodiment;

(7) FIG. **7** is a perspective view of a cart according to a second embodiment;

(8) FIG. **8** is a schematic sectional view of a recording device and the cart according to the second embodiment;

(9) FIG. **9** is a perspective view of a cart according to a third embodiment;

(10) FIGS. **10**A to **10**D are diagrams illustrating a roll setting procedure according to the third embodiment;

(11) FIG. **11** is a perspective view of a cart according to a fourth embodiment;

(12) FIGS. **12**A to **12**D are diagrams illustrating a roll setting procedure according to the fourth

embodiment;
(13) FIGS. **13**A and **13**B are views illustrating a state in which a paper-core fixing portion supports the roll;
(14) FIG. **14** is a perspective view illustrating a receiving portion of a cart according to a fifth embodiment;
(15) FIGS. **15**A and **15**B are perspective views illustrating a receiving portion of a cart according to a sixth embodiment;
(16) FIGS. **16**A to **16**C are a perspective view, a side view; and a top view of a cart according to a seventh embodiment;
(17) FIGS. **17**A and **17**B are a side view and a top view illustrating an accommodated form of a cart according to the seventh embodiment;
(18) FIGS. **18**A and **18**B are perspective views of a cart according to an eighth embodiment;
(19) FIG. **19** is a front view of the recording device;
(20) FIG. **20** is a schematic sectional view of the recording device illustrating a conveyance path of a sheet;
(21) FIGS. **21**A to **21**C are diagrams illustrating a state in which the roll is set on a roll holder;
(22) FIG. **22** is a perspective view of a roll set portion according to a ninth embodiment;
(23) FIG. **23** is a perspective view of a cart according to the ninth embodiment;
(24) FIGS. **24**A to **24**D are diagrams illustrating a roll setting procedure according to the ninth embodiment;
(25) FIGS. **25**A and **25**B are diagrams illustrating a space formed in the recording device according to the ninth embodiment;
(26) FIG. **26** is a side view of the recording device illustrating a movement path of a cart according to a tenth embodiment;
(27) FIGS. **27**A and **27**B are sectional views of a non-reference roll holder according to an eleventh of embodiment;
(28) FIG. **28** is a schematic side view of a recording device according to a twelfth embodiment;
(29) FIG. **29** is a perspective view of a periphery of a roll holder according to the twelfth embodiment;
(30) FIGS. **30**A and **30**B are perspective views illustrating an alignment adjustment mechanism according to the twelfth embodiment;
(31) FIG. **31** is a schematic side view of a recording device according to a thirteenth embodiment;
(32) FIG. **32** is a perspective view of a periphery of a roll holder according to the thirteenth embodiment; and
(33) FIG. **33** is a perspective view of a guide adjustment portion according to the thirteenth embodiment.

DESCRIPTION OF THE EMBODIMENTS

(34) Hereinafter, a description will be given, with reference to the drawings, of embodiments (examples) of the present invention. However, the sizes, materials, shapes, their relative arrangements, or the like of constituents described in the embodiments may be appropriately changed according to the configurations, various conditions, or the like of apparatuses to which the invention is applied. Therefore, the sizes, materials, shapes, their relative arrangements, or the like of the constituents described in the embodiments are not intended to limit the scope of the invention to the following embodiments.
(35) Recording devices which support rolls include a recording device which can handle a plurality of types of rolls with different widths. In a recording device of this type, rolls with various widths are supported by a holder by moving the holder in a width direction. However, in such a configuration that an insertion portion in a cantilever state is extended/contracted and holds the roll, when a roll with a particularly small width is to be supported, a protruding amount of the insertion portion becomes long and is easily deflected and thus, it is not appropriate to handle roll widths in a

wide range. Moreover, when the roll is disposed in the roll width direction with a shift with respect to an object support body, the roll interferes with an external device when it is transported by an object conveyance cart, and a position of the roll with respect to the object support body needs to be adjusted. A labor for adjusting the position of the roll with respect to the support body has poor workability particularly when a heavy roll is handled. Thus, first, in a work of mounting/removing the roll on/from the recording device while supporting the roll by the cart, a recording device, a cart, and a roll support mechanism, which widen a range of the roll widths that can be handled and can improve workability, will be described.

First Embodiment

(36) Recording Device

(37) First, a basic configuration of a recording device **100** which constitutes a roll attachment mechanism according to a first embodiment of the present invention will be described by referring to FIGS. **1** to **3**. As the recording device, an inkjet recording device including a sheet supply device which supplies a sheet as a print medium and a print portion which prints an image on the sheet will be described as an application example. Moreover, “ink” in this description is used as a collective name of liquids such as a recording liquid. Furthermore, when the same or corresponding members are provided in plural in the same figure of various devices which will be described below, suffixes such as a, b are added in illustration, but if there is no need for discrimination in the description, the suffixes such as a, b are omitted in description in some cases.

(38) FIG. **1** is a front view of the recording device **100**. The recording device **100** includes a roll set portion **200***a* on a paper-feed side and a roll set portion **200***b* on a take-up side, which can support a roll around which a sheet is wound in a roll state. The recording device **100** can print an image on the sheet fed from the roll supported by the roll set portion **200***a* and take up the printed sheet by the roll set portion **200***b*. A user can input various commands and the like such as size specification of the roll, setting of a roll type and the like to the recording device **100** by using various switches provided on a control panel. As shown in FIG. **1**, in the following description, a width direction of the roll supported by the recording device **100** is defined as an X-axis direction, a crossing direction crossing the width direction is a Y-axis direction, and a gravity direction as a Z-axis direction, respectively. In this embodiment, the X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to one another.

(39) FIG. **2** is a schematic sectional view of the recording device **100** illustrating a conveyance path of a sheet S. The sheet S pulled out of a roll R set on (attached to) the roll set portion **200***a* is connected to the roll set portion **200***b* and is taken up. Each of the roll set portions **200** includes a roll holder **2** and a holder guide **3** as a guide portion. The roll set portion **200**, the roll holder **2**, and the holder guide **3** provided on the paper-feed side and the take-up side, respectively, are given “a” at the end of the sign of the members on the paper-feed side and “b” at the end of the sign of the members on the take-up side and will be described by discriminating them as necessary.

(40) The sheet S pulled out of the roll R set on the roll set portion **200***a* is conveyed by an upstream sheet conveying portion **300** to a print portion **400** capable of printing an image. In the print portion **400**, the image is printed on the sheet S by ejecting the ink from an inkjet type printhead **8**.

(41) The printhead **8** ejects the ink from an ejection port by using an ejection energy-generating element such as an electricity-heat conversion element (heater), a piezo element and the like. When the electricity-heat conversion element is used, the ink is foamed by heat generation thereof, and the ink can be ejected from the ejection port by using the foaming energy. In an application of the present invention, the printhead **8** is not limited only to an inkjet type. Moreover, a print method of the print portion **400** is not limited, either, and it may be a serial scan type, a full-line type and the like, for example. In the case of the serial scan type, the image is printed with a conveying operation of the sheet S and scanning of the printhead **8** in a direction crossing the conveying direction of the sheet S. In the case of the full-line type, the printhead **8** which is lengthy and extends in the direction crossing the conveying direction of the sheet S is used, and an image is

printed while the sheet S is continuously conveyed.

(42) The roll R is held at a center part by the roll holder **2** and is rotated in arrow C**1**, C**2** directions by a roll drive motor, not shown. By enabling driving in both directions of forward rotation and reverse rotation, print can be performed selectively on a front surface or a back surface of the sheet S as shown by a path indicated by a solid line and a dotted line continuing from the roll set portion **200***a* on the paper-feed side in FIG. **2** to the upstream sheet conveying portion **300**. Similarly, the printed surface of the sheet S can be selectively taken up on an outer side or an inner side as shown by a path indicated by a solid line and a dotted line continuing from the downstream sheet conveying portion **500** in FIG. **2**, which will be described later, to the roll set portion **200***b* on the take-up side.

(43) An upstream conveyance guide **4** of the upstream sheet conveying portion **300** guides the sheet S to the print portion **400** while guiding the front surface or the back surface of the sheet S pulled out of the roll set portion **200***a*. A conveyance roller **5** is rotated forward or reversely by a conveyance-roller drive motor. A nip roller **6** is capable of being driven/rotated in accordance with rotation of the conveyance roller **5**. A conveying speed of the sheet S by the conveyance roller **5** is set higher than a pulling-out speed of the sheet S by the rotation of the roll R, whereby a back tension is given to the sheet S so that the sheet S can be conveyed in a state in which a tension remains to be applied to the sheet S. As a result, sagging of the sheet S is prevented, and occurrence of a crease in the sheet S and occurrence of a conveyance error can be suppressed.

(44) A platen **7** in the print portion **400** adsorbs a surface on a side opposite to a recording surface of the sheet S through a suction hole **7***a* by a negative pressure generated by a suction fan **9**. As a result, by regulating the position of the sheet S so that it goes on the platen **7**, high-precision print of an image by the printhead **8** is made possible.

(45) A downstream conveyance guide **10** of the downstream sheet conveying portion **500** leads the sheet S to the roll set portion **200***b* while guiding the front surface or the back surface of the sheet S pulled out of the print portion **400**. By fixing a distal end of the sheet S to a paper core which is set on the roll set portion **200***b* and by rotating the set paper core in accordance with the conveying speed of the conveyance roller **5**, the sheet S printed by the print portion **400** can be continuously taken up.

(46) The recording device **100** includes a stand **14** on both end parts in the width direction, respectively. The stand **14** supports weights of the upstream sheet conveying portion **300**, the print portion **400**, and the downstream sheet conveying portion **500** on an upper part of a main body and supports the weight of the roll R through the roll set portion **200**. The stand **14** is in contact with a floor surface and releases the weight of the recording device **100** to the floor surface.

(47) FIG. **3** is a perspective view of the roll set portion **200** of the recording device **100** according to a first embodiment. Into a hollow hole portion of the paper core of the roll R, a shaft-shaped paper-core fixing portion **11** provided on the roll holder **2** is inserted. The paper-core fixing portion **11** is rotated forward or reversely and driven by a roll drive motor, not shown. The roll holder **2** provided on the roll set portion **200** is constituted by a reference roll holder **21** and a non-reference roll holder **22**, which make a pair as a roll support portion. On each of the reference roll holder **21** and the non-reference roll holder **22**, the paper-core fixing portion **11** is rotatably mounted. The non-reference roll holder **22** is held slidably in the width direction (X-direction) of the roll R by the holder guide **3**. Moreover, in the non-reference roll holder **22**, a lock mechanism **12** is provided, and by turning a knob provided in the lock mechanism **12** in a T-direction, the non-reference roll holder **22** can be fixed at an arbitrary position in the width direction with respect to the holder guide **3**. Though the reference roll holder **21** is also held by the holder guide **3**, a slide mechanism is not indispensable. A friction member is provided in the paper-core fixing portion **11**, and when the roll R is set, the roll R and the paper-core fixing portion **11** are integrally rotated.

(48) By holding the roll R by using the slide mechanism by the roll holder **2** and the holder guide **3**, the roll R is firmly supported by the roll set portion **200** regardless of the roll width. That is, the

recording device **100** is so configured that the roll R with an arbitrary width not larger than a width-direction length of the holder guide **3** can be set. Moreover, if it is so configured that the reference roll holder **21** is fixed to the holder guide **3** so as not to slide, the end part of the roll R is fitted with the fixed reference roll holder **21** and thus, it is configured such that a reference-side end-part position of the roll R is determined constant with respect to the roll set portion **200**.
(49) The holder guide **3** of this embodiment is fixed to the stand **14**. Moreover, a roll holder **2***a* on the paper-feed side is configured to be disposed on a front side of the recording device (Y-positive direction) with respect to the holder guide **3***a*. That is, the holder guide **3***a* is disposed on a main-body depth side (Y-negative direction) with respect to the paper-core fixing portion **11**. By configuring as above, since there is no structure such as the holder guide **3** on a main-body front side or on a main-body lower side of the paper-core fixing portion **11** with respect to the roll holder **2***a* on the paper-feed side, there is no obstacle in the mounting/removing work of the roll R. Therefore, in the recording device **100**, by placing the roll R on the cart and moving the cart from a P-direction shown in FIG. **3**, the roll R can be set. The recording device **100** further includes an abutting surface **15** as a positioning portion which determines a position of the cart in the crossing direction (Y-direction), which is located lower than the holder guide **3** and extends between the stands **14**. Moreover, the roll holder **2***b* on the take-up side is configured to be disposed on a back-surface side of the recording device (Y-negative direction) with respect to the holder guide **3***b*, and workability of mounting/removing of the roll R is favorable. Details of a positioning method in the crossing direction of the cart will be described later.
(50) Cart
(51) Subsequently, the cart **50** according to the first embodiment, which supports and conveys the roll R when the roll R is mounted on the roll holder **2**, will be described by referring to FIG. **4**. FIG. **4** is a perspective view of the cart **50** according to the first embodiment. The cart **50** includes a receiving portion **51** which supports the roll R, a wheel portion **52**, a height adjustment portion **53** which adjusts a height of the receiving portion **51**, a cart base portion **55** which supports the receiving portion **51** through the height adjustment portion **53**, and an abutting portion **54** which protrudes in the crossing direction from the cart base portion **55**.
(52) The receiving portion **51** as a roll receiving portion can support the roll R from below in a state in which the width direction of the roll R is horizontal, that is, with the same attitude as that of the roll R, when the roll IR is mounted on the roll holder **2**. The receiving portion **51** has receiving-portion inclined surfaces **51***a*, **51***b* as two inclined surfaces in contact with the roll R from below. When the receiving portion **51** is viewed from the width direction, the two receiving-portion inclined surfaces **51***a*, **51***b* form a V-shape. In a state in which the roll R is placed on the receiving portion **51**, the roll R is in contact with both the receiving-portion inclined surfaces **51***a*, **51***b*, and the receiving-portion inclined surfaces **51***a*, **51***b* are located symmetrically to a center of the roll R. That is, even if a winding diameter of the roll R is different, a position in the crossing direction (Y-direction) of the circumference center of the roll R to the cart **50** is the same position at all times.
(53) The cart **50** is freely movable on the floor by the wheel portion **52** provided below the cart base portion **55**. The height adjustment portion **53** provided on a lower side of the receiving portion **51** is constituted so as to be capable of extension/contraction in a vertical direction (Z-direction) and elevates up/down the receiving portion **51**. The abutting portion **54** is brought into contact with the abutting surface **15** between the stands **14** when the cart **50** on which the roll R is placed is inserted into the recording device **100** and determines a cross-direction position of the cart **50** with respect to the stands **14**.
(54) Roll Attaching Method
(55) Subsequently, details of a method of mounting the roll Ron the roll holder **2** by using the cart **50** will be described by referring to FIGS. **5**A to **5**D, and **6**. The roll attachment mechanism according to the present invention is constituted by the recording device and the cart. In the following, a mounting/removing work of the roll R to/from the roll holder **2***a* on the paper-feed

side will be described as an example, but the roll R can be mounted on/removed from the roll holder **2***b* on the take-up side with the similar method.
(56) FIGS. **5**A to **5**D are front views illustrating a procedure of setting the roll on the roll set portion **200** by using the cart **50** in the first embodiment. FIG. **5**A illustrates a state of positioning of the roll R in the vertical direction after the cart **50** has been moved to a space between the opposing paper-core fixing portions **11**. FIG. **5**B illustrates a state in which the cart **50** on which the roll R is placed is moved to the reference roll holder **21** side. FIG. **5**C illustrates a state in which the non-reference roll holder **22** is moved to the roll R side while one end in the width direction of the roll R is in contact with the paper-core fixing portion **11**. FIG. **5**D illustrates a state in which the non-reference roll holder **22** is fixed by the lock mechanism **12**, and the support of the roll R is completed. FIG. **6** is an A-A sectional view in FIG. **5**A and is a schematic sectional view illustrating a positional relationship of the recording device **100** and the cart **50**.
(57) When the roll R placed on the receiving portion **51** of the cart **50** is to be set on the roll set portion **200**, first, the cart **50** is moved so that the roll R is positioned in a region sandwiched by the two roll holders **2**. Then, as shown in FIG. **6**, by bringing the abutting portion **54** of the cart **50** into contact with the abutting surface **15**, the crossing-direction (Y-direction) position of the cart **50** with respect to the roll set portion **200** is determined. That is, the abutting portion **54** as the positioned portion is brought into contact with the abutting surface **15** as the positioning portion, whereby the position in the crossing direction of the abutting portion **54** is determined with respect to the abutting surface **15**. Moreover, since movement in the width direction (X-direction) of the abutting portion **54** to the abutting surface **15** is not limited, the wheel portion **52** functions as the position adjustment mechanism of the roll, and the cart **50** can be moved in the width direction with the roll R supported while being positioned in the crossing direction. Therefore, the position in the width direction of the roll R can be adjusted while the roll R is positioned in the crossing direction with respect to the recording device **100**.
(58) The abutting surface **15** is provided in both directions of the paper-feed side and the take-up side with respect to the stands **14** and extends in the region in the width direction between the two stands **14**. Therefore, as long as the cart **50** is located between the two stands **14**, the abutting portion **54** can be brought into contact with the abutting surface **15** regardless of the position in the width direction. It is configured such that, when the abutting portion **54** is brought into contact with the abutting surface **15**, a distance from a contact surface of the abutting portion **54** to the center of the roll R and a distance from the abutting surface **15** to a center of the paper-core fixing portion **11** have the same length in the crossing direction. In other words, in a state in which the abutting portion **54** is in contact with the abutting surface **15**, positions in the crossing direction of the center of the roll R and the center of the paper-core fixing portion **11** match each other. Moreover, since the receiving portion **51** is configured such that the crossing-direction position of the center of the roll R with respect to the cart **50** is the same position at all times regardless of the winding diameter of the roll R, the crossing-direction position of the roll R with respect to the paper-core fixing portion **11** can be determined without being influenced by the winding diameter.
(59) After the crossing-direction position of the roll R is determined by the positioning portion and the positioned portion, as in FIG. **5**A, the height of the center of the roll R and the height of the paper-core fixing portion **11** in the recording device **100** are matched with each other, and positioning in the vertical direction is performed. The height adjustment portion **53** is provided on the cart **50**, and the height of the roll R (Z-direction position) is adjusted by using this. Subsequently, as in FIG. **5**B, the one end of the paper core of the roll R is inserted into the paper-core fixing portion **11** of the reference roll holder **21** by moving the cart **50** in the width direction, Since the center of the roll R matches the center of the paper-core fixing portion **11** both at the crossing-direction position and the vertical-direction position by the operations so tar, the paper core can be easily inserted into the paper-core fixing portion **11**. When the one end of the paper core of the roll R is brought into contact with the paper-core fixing portion **11**, the friction member

of the paper-core fixing portion **11** is fitted in the paper core of the roll R. As a result, the roll R is fixed to the paper-core fixing portion **11** of the reference roll holder **21**. Moreover, as in FIG. **5**C, by sliding the non-reference roll holder **22** in the width direction so as to get closer to the roll R, the other end of the paper core of the roll R is inserted into the paper-core fixing portion **11** of the non-reference roll holder **22**. As a result, similarly to the reference roll holder **21**, the roll IR is fixed to the paper-core fixing portion **11** of the non-reference roll holder **22**, and the width-direction position of the roll R is determined. Lastly, as in FIG. **5**D, the non-reference roll holder **22** is fixed to the holder guide **3** by using the lock mechanism **12**. As a result, sliding of the non-reference roll holder **22** in the paper width direction, which causes removal of the roll R from the roll holder **2**, is prevented. Subsequently, by elevating the receiving portion **51** downward and by retreating it from the roll R by the height adjustment portion **53** of the cart **50**, setting of the roll R on the recording device **100** is completed.

(60) The setting procedure of the roll R on the roll set portion **200** has been described so far, and by performing this setting procedure in a reverse order, the roll R can be removed from the roll set portion **200**. By configuring such that the cart can be used for the removal of the roll R, operations from the removal of the roll from the recording device to the conveyance of the roll R can be performed without unloading the roll R from the cart.

(61) As described above, according to the roll attachment mechanism according to this embodiment, the operations from the conveyance of the roll to the mounting of the roll on the recording device or from the removal of the roll from the recording device to the conveyance can be performing without unloading the roll from the cart even once. When the roll is mounted/removed, the weight of the roll is always supported by the cart and thus, the roll can be mounted/removed easily without manually supporting the weight of the roll. Moreover, by means of the slide mechanism of the roll holder and the cart which is freely movable in the width direction even in the state of being positioned in the crossing direction, the configuration can deal with a wide range of roll widths. Moreover, during the mounting/removing works of the roll, the cart can be moved in the width direction in the state in which the cart is positioned in the crossing direction. Therefore, when the roll is to be loaded on the cart, a degree of freedom in the width-direction position of the roll with respect to the cart is high, and there is no need to manually position the heavy roll and thus, workability of mounting/removal of the roll is favorable.

Second Embodiment

(62) Subsequently, a second embodiment will be described by referring to FIGS. **7** and **8**. In the second embodiment, a method of determining the crossing-direction position of the cart with respect to the recording device is different from that of the first embodiment. Note that a difference from the first embodiment is mainly described in the following, while the other identical configurations are given the same signs, and the description thereof is omitted.

(63) FIG. **7** is a perspective view of a cart **250** in the second embodiment. FIG. **8** is a schematic sectional view when the cart **250** and a recording device **600** are viewed in the positional relationship similar to that in FIG. **6**. The cart **250** includes a coupling lever **56** and a swing shaft **59** as a lever mechanism, and a coupling portion **57** is disposed at a distal end of the coupling lever **56**. The coupling lever **56** is swingable around the swing shaft **59**. Moreover, between the two stands **14**, a coupling rail **16** extending in the width direction (X-direction) is provided. The recording device **600** and the cart **250** are coupled by fitting of the coupling portion **57** in the coupling rail **16**, and the crossing-direction (Y-direction) position of the cart **250** with respect to the recording device **600** is determined. The coupling lever **56** can be changed to a first attitude fitted with the coupling rail **16** and a second attitude in which the fitting with the coupling rail **16** is released by swinging around the swing shaft **59**. Moreover, in this embodiment, when the position of the coupling portion **57** in the crossing direction is determined by the coupling rail **16**, movement of the cart **250** in a direction separated away from the recording device **600** is also limited.

(64) The coupling rail **16** extends to a region between the two stands **14** in the width direction. Therefore, as long as the cart **250** is located between the two stands **14**, the cart **250** can be coupled with the recording device **600** regardless of the position in the width direction. Here, in the crossing direction, it is so configured that a distance from the coupling portion **57** to the center of the roll R and a distance from the coupling rail **16** to the center of the paper-core fixing portion **11** have the same length. That is, in the state in which the coupling portion **57** and the coupling rail **16** are coupled, the crossing-direction positions of the center of the roll R and the center of the paper-core fixing portion **11** match each other. That is, in this embodiment, the coupling rail **16** functions as the positioning portion, and the coupling portion **57** as the positioned portion.
(65) In the second embodiment, the procedure of setting the roll R on the roll set portion **200** by using the cart **250** and the procedure of removing the roll R from the roll set portion **200** are similar to FIGS. **5**A to **5**D illustrated in the description of the first embodiment.
(66) When the roll R placed on the receiving portion **51** of the cart **250** is to be set on the roll set portion **200**, first, the cart **250** is moved so that the roll R is positioned in a region sandwiched by the two roll holders **2**. By moving the cart **250** in the crossing direction so as to get closer to the recording device **600**, the coupling lever **56** swings around the swing shaft **59**, the coupling portion **57** is fitted in the coupling rail **16**, and the crossing-direction position of the cart **250** with respect to the recording device **600** is determined. As described above, in the second embodiment, by causing the coupling portion **57** to be coupled with the coupling rail **16**, the cart **250** can be freely moved in the width direction while the crossing-direction position of the cart **250** with respect to the recording device **600** is fixed. That is, the cart **250** can be moved in the width direction while the state in which the crossing-direction positions of the center of the roll R and the center of the paper-core fixing portion **11** match each other is maintained.
(67) As described above, according to this embodiment, in the state in which the crossing-direction position of the roll with respect to the recording device is determined, the roll can be mounted on the roll holder which is slidable in the width direction by adjusting the width-direction position of the roll easily by the position adjustment mechanism. Therefore, in the mounting/removing work of the roll, a wide range of the roll widths can be handled, and the workability is improved.
Third Embodiment
(68) Subsequently, a third embodiment will be described by referring to FIGS. **9**, and **10**A to **10**D. In the third embodiment, when the roll is to be se on the recording device, a method of inserting a paper core of the roll in the paper-core fixing portion is different from that of the first embodiment. Note that a difference from the first embodiment is mainly described in the following, while the recording device **100** and other identical configurations are given the same signs, and the description thereof is omitted.
(69) FIG. **9** is a perspective view of a cart **350** in the third embodiment. As shown in FIG. **9**, the cart **350** in this embodiment includes a slider **60** on a lower part of the receiving portion **51**, By means of the slider **60** as the position adjustment mechanism of the roll R, the receiving portion **51** can slide in the width direction (X-direction) with respect to the wheel portion **52** and the cart base portion **55**.
(70) FIGS. **10**A to **10**D are front views illustrating a procedure of setting the roll R on the roll set portion **200** by using the cart **350** in the third embodiment. FIG. **10**A illustrates a state in which the cart **350** is moved to the opposing paper-core fixing portions **11**, and the roll R is positioned in the vertical direction (Z-direction). FIG. **10**B illustrates a state in which the roll R positioned in the vertical direction and the receiving portion **51** are moved by the slider **60** to the reference roll holder **21** side. FIG. **10**C illustrates a state in which the non-reference roll holder **22** is moved to the roll R side in a state in which the one end in the width direction of the roll R is in contact with the paper-core fixing portion **11**. FIG. **10**D illustrates a state in which the non-reference roll holder **22** is fixed by the lock mechanism **12**, and the support of the roll R is completed.
(71) A point that, in order to determine a position in the crossing direction (Y-direction) of the cart

350 with respect to the recording device **100**, the abutting portion **54** is brought into contact with the abutting surface **15** is similar to the first embodiment. After that, as in FIG. **10**A, the height of the center of the roll R is matched with the height of the paper-core fixing portion **11** in the recording device **100**, and the positioning, in the vertical direction is completed. Subsequently, as in FIG. **10**B, by moving the receiving portion **51** on Which the roll R is placed in the width direction by the slider **60** of the cart **350**, the one end of the paper core of the roll R is inserted into the paper-core fixing portion **11** of the reference roll holder **21**. In the operations so far, the center of the roll R matches the center of the paper-core fixing portion **11** at both the crossing-direction position and the vertical-direction position and thus, the paper core can be easily inserted to the paper-core fixing portion **11**. A method of inserting the other end of the paper core of the roll R into the paper-core fixing portion **11** of the non-reference roll holder **22** so as to complete the setting of the roll R after that is similar to the first embodiment as shown in FIGS. **10**C and **10**D.
(72) As described above, according to this embodiment, only the receiving portion **51** can be moved by the slider **60** without moving the entire cart **350** in the width direction and thus, the roll R can be moved in the width direction with a smiler three as compared with the first embodiment, That is, since the position in the width direction of the roll can be adjusted more easily by the position adjustment mechanism in a state in which the roll is positioned in the crossing direction with respect to the recording device, the workability of mounting/removal of the roll is further improved.
Fourth Embodiment
(73) Subsequently, a fourth embodiment will be described by referring to FIGS. **11**, and **12**A to **12**D. In the fourth embodiment, when the roll is to be set on the recording device, a method of inserting a paper core of the roll into the paper-core fixing portion is different from the first embodiment and the third embodiment. Note that a difference from the first embodiment is mainly described in the following, while the recording device **100** and other identical configurations are given the same signs, and the description thereof is omitted.
(74) FIG. **11** is a perspective view of a cart **450** in the fourth embodiment. As shown in FIG. **11**, the cart **450** in this embodiment includes a receiving-portion roller **61** on an end part in a longitudinal direction (X-direction) of the receiving-portion inclined surfaces **51***a*, **51***b* of a receiving portion **451**. The receiving-portion roller **61** as the position adjustment mechanism of the roll is rotatable in a state in which the roll R is placed thereon. By rolling of the receiving-portion roller **61** in contact with the roll R, the roll R can be moved in the width direction with a small force. The receiving-portion roller **61** extends from a vicinity of upper ends of the receiving-portion inclined surfaces **51***a*, **51***b* to a vicinity of a lower end so that it can contact the roll R compliant with winding diameters of the various rolls R.
(75) FIGS. **12**A to **12**D are front views illustrating a procedure of setting the roll R on the roll set portion **200** by using the cart **450** in the fourth embodiment. FIG. **12**A illustrates a state in which the cart **450** is moved to a space between the opposing paper-core fixing portions **11**, and the roll R is positioned in the vertical direction (Z-direction). FIG. **12**B illustrates a state in which the roll R on the receiving-portion roller **61** positioned in the vertical direction is moved to the reference roll holder **21** side. FIG. **12**C illustrates a state in which the non-reference roll holder **22** is moved to the roll IR side in a state in which the one end in the width direction of the roll R is in contact with the paper-core fixing portion **11**. FIG. **12**D illustrates a state in which the non-reference roll holder **22** is fixed by the lock mechanism **12**, and the support of the roll R is completed.
(76) A point that, in order to determine a position in the crossing direction (Y-direction) of the cart **450** with respect to the recording device **100**, the abutting portion **54** is brought into contact with the abutting surface **15** is similar to the first embodiment. After that, as in FIG. **12**A, the height of the center of the roll R is matched with the height of the paper-core fixing portion **11** in the recording device **100**, and the positioning in the vertical direction is completed. Subsequently, as in FIG. **12**B, by moving the roll R in the width direction, and the one end of the paper core of the roll

IR is inserted into the paper-core fixing portion **11** of the reference roll holder **21**. A method of inserting the other end of the paper core of the roll R into the paper-core fixing portion **11** of the non-reference roll holder **22** so as to complete the setting of the roll R after that is similar to the first embodiment as shown in FIGS. **12**C and **12**D.

(77) As described above, according to this embodiment, only the roll R can be moved in the width direction by the receiving-portion roller **61** without moving the entire cart **450** in the width direction and thus, the roll R can be moved in the width direction with a smaller force as compared with the first embodiment. That is, since the position in the width direction of the roll can be adjusted more easily by the position adjustment mechanism in a state in which the roll is positioned in the crossing direction with respect to the recording device, the workability of mounting/removal of the roll is further improved.

Fifth Embodiment

(78) Subsequently, a fifth embodiment will be described by referring to FIGS. **13**A, **13**B, and **14**, The fifth embodiment is an effective configuration when a roll with a width smaller than a width in a longitudinal direction of the receiving portion of the cart is set. Note that a difference from the first embodiment is mainly described in the following, while the recording device **100** and other identical configurations are given the same signs, and the description thereof is omitted.

(79) FIGS. **13**A and **13**B are diagrams illustrating a state in which the paper-core fixing portion **11** supports the roll R with a different paper-core diameter. FIG. **13**A illustrates a state in which a roll RA including a paper core **81**A with an inner diameter DA is supported, and FIG. **13**B illustrates a state in which a roll RB including a paper core **81**B with an inner diameter DB larger than the inner diameter DA is supported. A size of the roll R handled by the recording device **100** of the present invention is assumed to be a diameter of approximately 200 mm at the maximum, a width of 1630 mm, a weight of approximately 50 kg, and 2 inches/3 inches of a paper core size. The paper-core fixing portion **11** in this embodiment includes regions engaged with an inner diameter of the paper cores of 2 inches and 3 inches, respectively, in an outer diameter part of the paper-core fixing portion **11** which is inserted and fixed to the paper core of the roll R and supports the roll R in order to handle different paper-core sizes. When the roll R with the paper-core size of 2 inches is to be supported, the paper-core fixing portion **11** supports the roll R in the state shown in FIG. **13**A, and in the case of supporting the roll R with the paper-corer size of 3 inches, in the state shown in FIG. **13**B.

(80) In order to stably support the roll R by the receiving portion **51** of the cart **50**, the roll R is supported to the end part in the width direction by the receiving portion **51** so that a contact area with the roll R is increased and thus, the length of the receiving portion **51** is preferably as long as possible. However, if the roll width is small with respect to the receiving portion **51**, when the roll with a paper-core size of 2 inches and a small winding diameter is to be mounted, the paper-core size of the paper-core fixing portion **11** might interfere with the receiving portion **51** by the part which supports the roll R with 3 inches. Particularly, when the roll R whose sheet has been consumed to the vicinity of the winding core and the winding diameter is reduced is to be mounted, the paper-core fixing portion **11** can easily interfere with the receiving portion **51**. Then, since the roll R cannot be set appropriately by the paper-core fixing portion **11**, prolongation of the length of the receiving portion **51** larger than the roll width is not preferable, either. In this embodiment, since the roll width mainly used in the market is 600 mm or more, the length in the longitudinal direction of the receiving portion **51** is 600 mm.

(81) However, there are extremely small rolls with the roll widths less than 600 mm in the market. In the cart **50** shown in the first embodiment, for example, when the extremely small roll with the roll width of 200 mm and the paper-core size of 2 inches is to be mounted on the receiving portion **51** whose length in the longitudinal direction is 600 mm, the paper-core fixing portion **11** can interfere with the receiving portion **51**. Thus, as the fifth embodiment, a configuration of the cart that can also handle the extremely small roll will be described.

(82) FIG. **14** illustrates a receiving portion **551** of the cart according to the fifth embodiment. A length L1 in the width direction (X-direction), which is the longitudinal direction of the receiving portion **551**, is 600 mm, and a length L2 in the crossing direction (Y-direction), which is a short-side direction orthogonal to the longitudinal direction, is 200 mm. The receiving portion **551** has the receiving-portion inclined surfaces **51***a*, **51***b* extending in the width direction and receiving-portion inclined surfaces **62***a*, **62***b*, **63***a*, **63***b* extending in the crossing direction. By means of these receiving-portion inclined surfaces, the receiving portion **551** includes a part looking like a V-shape when seen from the width direction and a part looking like a V-shape when seen from the crossing direction.
(83) By means of the aforementioned configuration, when the roll width is 600 mm or more, the roll R is installed on the receiving portion **551** so that the width direction of the roll R matches a width direction of the receiving portion **551**, and the roll R can be supported by the receiving-portion inclined surfaces **51***a*, **51***b*. On the other hand, when the roll width is 200 mm or more and less than 600 nm, the roll R is installed on the receiving portion **551** so that the width direction of the roll R matches the crossing direction of the receiving portion **551**, and the roll R can be supported by the receiving-portion inclined surfaces **62***a*, **62***b* or the receiving-portion inclined surfaces **63***a*, **63***b*. That is, when the roll width is 200 mm or more and less than 600 mm, the direction of the cart with respect to the roll R and the recording device **100** at the roll set are rotated by 90° as compared with the case where the roll width is 600 mm or more. As described above, by installing the roll R on the cart by changing the direction of the roll R with respect to the cart depending on the roll width, the cart can support the roll R in a state in which the receiving portion of the cart is smaller than the width of the roll. Therefore, even if the extremely small roll is to be mounted on the recording device **100**, the roll R can be firmly supported without interference by the receiving portion **551** with the paper-core fixing portion **11**.
(84) Moreover, in the cart of this embodiment, the positioned portion with respect to the recording device **100** such as the abutting portion and the like is provided on the end part in the width direction (longitudinal direction) and the both end parts in the crossing direction (short-side direction) of the cart. By providing the positioned portion on the end parts in the width direction and the crossing direction of the cart, respectively, even when a direction of installing the roll R on the cart is changed as described above, the cart can be positioned in the crossing direction with respect to the recording device **100**.
(85) As described above, according to this embodiment, the roll with a roll width smaller than that of the first embodiment can be set on the recording device in a state supported by the cart. Therefore, a range of the roll widths that can be handled can be further widened.
Sixth Embodiment
(86) Subsequently, a sixth embodiment will be described by referring to FIGS. **15**A and **15**B. The sixth embodiment is a configuration which is effective when a roll with a width smaller than a width in the longitudinal direction of the receiving portion of the cart is set and is a mode different from the fifth embodiment. Note that a difference from the fifth embodiment is mainly described in the following, while the recording device **100** and other identical configurations are given the same signs, and the description thereof is omitted.
(87) FIG. **15**A illustrates a receiving portion **651** of the cart according to the sixth embodiment. The receiving portion **651** has a divided receiving portions **64**, **65**, which are divided into two parts in the width direction (X-direction). A length L3 in the width direction of the receiving portion **651** is 600 mm, and a length IA in the width direction of the divided receiving portion **64** is 200 nm. The divided receiving portions **64**, **65** have the receiving-portion inclined surfaces **51***a*, **51***b* extending in the width direction, respectively. When the roll width is 600 nm or more, the roll R can be supported at two spots or more by placing the roll R across the divided receiving portions **64**, **65** and thus, it can be firmly supported.
(88) FIG. **15**B illustrates a state in which the divided receiving portion **64** of the cart according to

the sixth embodiment is rotated. The divided receiving portion **64** can be rotated by 90° around a rotation axis (not shown) extending in the vertical direction from the state in FIG. **15**A. When the divided receiving portion **64** is rotated by 90°, the receiving-portion inclined surfaces **51***a*, **51***b* of the divided receiving portion **64** extending in the width direction extends in the crossing direction (Y-direction). In the case where the roll width is 200 mm or more and less than 600 mm, by installing the roll R on the divided receiving portion **64** in the state in which the divided receiving portion **64** shown in FIG. **15**B is rotated, the width of the divided receiving portion **64** becomes the roll width or less. Therefore, according to the cart according to this embodiment, the roll R can be set on the recording device without interference by the paper-core fixing portion **11** with the divided receiving portion **64**. Moreover, since the roll R can be set on the recording device **100** by changing the direction of the cart with respect to the recording device **100**, interference by the cart with the recording device can be also prevented.

(89) Moreover, in the cart of this embodiment, the positioned portion with respect to the recording device **100** such as the abutting portion or the like is provided on the end, part in the width direction (longitudinal direction) of the cart and the end part in the crossing direction (short-side direction) of the cart. By providing the positioned portion on the end parts in the width direction and the crossing direction of the cart, respectively, even when the direction in which the roll R is installed on the cart is changed as described above, the cart can be positioned in the crossing direction with respect to the recording device **100**.

(90) As described above, according to this embodiment, the roll with the roll width smaller than that in the first embodiment can be set on the recording device in the state of being supported by the cart. Therefore, a range of the roll widths that can be handled can be further widened.

(91) In this embodiment, the divided receiving portion **64** is configured to be rotatable, but the divided receiving portion **65** may also be configured to be rotatable. Moreover, by setting the lengths in the longitudinal direction of the divided receiving portions **64**, **65** different from each other, each of the divided receiving portions may be configured to be used separately in accordance with the roll width. At that time, the positioned portion with respect to the recording device is preferably provided on both end parts in the width direction of the cart.

Seventh Embodiment

(92) Subsequently, a seventh embodiment will be described by referring to FIGS. **16**A to **16**C, **17**A, and **17**B. A cart **750** according to the seventh embodiment is a configuration in which an accommodation space when not in use is reduced as compared with the first embodiment. Note that a difference from the first embodiment is mainly described in the following, while the recording device **100** and other identical configurations are given the same signs, and the description thereof is omitted.

(93) After the roll R is set on the roll set portion **200** by using the cart, the worker needs to manually operate the roll R so that the sheet can be fed from the roll R. When the roll R is operated, the receiving portion or a handle portion (not show) of the cart located in the periphery of the roll R becomes an obstacle in a state in which the cart is engaged with the recording device, which obstructs the work. Therefore, after the roll R is set on the roll set portion **200**, the cart is preferably removed from the recording device **100** and moved to a remote position.

(94) Moreover, in the recording device in which the roll holder is provided in a paper feed portion and a take-up portion, respectively, a plurality of workers perform a transport work and the like of the roll R for paper-feed and a product roll taken up after print at the same time in parallel and thus, the carts are preferably prepared in plural. When the mounting and removal of the rolls are performed at the same time in the plurality of recording devices, too, a more efficient operation is similarly enabled by preparing a plurality of the carts, in view of the circumstances described above, the seventh embodiment will be described below as a mode of a cart capable of accommodation by reducing an occupied space when a plurality of the carts are stored in one spot.

(95) FIG. **16**A is a perspective view of a cart **750** according to the seventh embodiment. By means

of the cart **750**, as compared with the cart **50** according to the first embodiment, a grounding area when the cart is not in use can be reduced for storage. The receiving portion **51** of the cart **750** has a configuration similar to that the first embodiment and includes the receiving-portion inclined surfaces **51***a*, **51***b* in contact with the roll R. On the other hand, the cart **750** is different from the cart **50** according to the first embodiment in a shape of a cart base portion **755** which supports the height adjustment portion **53** from below. The cart base portion **755** includes a notch **755***a* on one end in the crossing direction (Y-direction) and at the center in the width direction (X-direction). Moreover, the cart base portion **755** includes a notch **755***b* also on the other end on a side opposite to the notch **755***a* in the crossing direction and on both ends in the width direction. Furthermore, on a lower surface of the cart base portion **755**, a front wheel portion **52***a*, a rear wheel portion **52***b*, and a protruding portion **68** are provided. The cart **750** is freely movable on the floor by the front wheel portion **52***a*, and the rear wheel portion **52***b*. The protruding portion **68** protrudes downward (Z-positive direction) from the cart base portion **755** and extends in the width direction (Y-direction). Moreover, at a distal end of the protruding portion **68**, a taper shape made of two inclined surfaces is provided.

(96) FIG. **16**B illustrates the cart **750** when seen from the width direction (X-negative direction). A protruding amount of the protruding portion **68** is such a degree that a distal end of the protruding portion **68** does not protrude farther than a grounding part of the wheel portion, and the protruding portion **68** does not interfere with traveling of the cart **750**. Moreover, the protruding portion **68** is located immediately below the receiving portion **51**. The reason why such a positional relationship was taken will be described later.

(97) A width in the crossing direction of the receiving portion **51** is such a degree that is approximately the same as the winding diameter of the roll R when a sheet residual amount is small so that the roll R can be unloaded easily even when the winding diameter of the roll R is small. On the other hand, when the cart **750** is seen from the width direction, the front wheel portion **52***a* and the rear wheel portion **52***b* are located on an outer side of the receiving portion **51**. In this embodiment, an interval in the crossing direction between the wheel portions **52** is approximately the same as the winding diameter of the roll R When the residual amount of the roll R is large, and they are in such a positional relationship with respect to the receiving portion **51** that the wheel portions **52** protrude to both sides in the crossing direction, respectively. By means of such disposition, even the roll R with a large sheet residual amount and a large winding diameter can be stably transported by the cart **750**.

(98) FIG. **16**C illustrates the cart **750** when seen from above (Z-negative direction). A length in the width direction of the receiving portion **51** corresponds to various roll widths and thus, it is determined by considering a roll with the smallest roll width in the rolls R assumed to be supported. By setting the length in the width direction of the receiving portion **51** smaller than the smallest roll width, operability in mounting/removing of the roll R is not impaired by the receiving portion **51**. On the other hand, when the cart **750** is seen from the vertical direction (Z-direction), the front wheel portion **52***a* and the rear wheel portion **52***b* are located on the outer side of the receiving portion **51** in the width direction. In this embodiment, the interval in the width direction between each of the front wheel portion **52***a* and the rear wheel portion **52***b* is approximately the same as the largest roll width in the roll widths assumed to be supported, and they are in such a positional relationship that the wheel portion **52** protrudes on both sides in the width direction with respect to the receiving portion **51**. By means of such disposition, even the roll R with a large width can be stably transported by the cart **750**.

(99) As described above, in the cart base portion **755**, a notch **755***a* is provided on one end side in the crossing direction and a notch **755***b* on the other end side. The notch **755***a* is located at a center part in the width direction of the cart base portion **755**, and on both end parts in the width direction of the cart base portion **755**, a protruding portion **755***c* protruding on the one end side in the crossing direction is formed. The notch **755***b* is located on both ends in the width direction of the

cart base portion **755**, and a protruding portion **755***d* protruding to the other end side in the crossing direction is formed at the center part in the width direction of the cart base portion **755**. A length X1 in the width direction of the notch **755***a* on the one end side is larger with respect to a length X2 in the width direction of the protruding portion **755***d* on the other end side, and a relationship of X1>X2 is satisfied. Moreover, when seen from the crossing direction, the notch **755***a* and the protruding portion **755***d* overlap each other, and the notch **755***b* and the protruding portion **755***c* overlap each other. The protruding portion **755***c* and the protruding portion **755***d* may have a function as the positioned portions in contact with the positioning portion of the recording device **100**.

(100) The front wheel portion **52***a* is provided on the protruding portion **755***c* on the one end side in the crossing direction of the cart base portion **755**, and the rear wheel portion **52***b* is provided on the protruding portion **755***d* on the other end side in the crossing direction of the cart base portion **755**. That is, the front wheel portion **52***a* and the rear wheel portion **52***b* are provided at shifted positions in the crossing direction.

(101) Subsequently, an accommodation-space reduction effect when a plurality of the carts **750** are accommodated will be described. FIG. **17**A illustrates a state in which a plurality of the carts **750** stacked vertically are seen from the width direction (X-negative direction). When the cart **750** is to be accommodated, as shown in FIG. **17**A, a second cart **750**B can be stacked on a first cart **750**A. At this time, a tapered surface of the receiving portion **51** provided on the upper part of the first cart **750**A is engaged with the tapered surface of the protruding portion **68** provided on the lower part of the second cart **750**B. By configuring as above, the first cart **750**A can stably support the second cart **750**B. Moreover, another cart can be also stacked on the second cart **750**B. By having the configuration that the carts can be stably stacked vertically as above, a grounding area for two units or more of the carts can be an area for one unit of the cart and thus, an occupied space when the cart is not in use can be reduced.

(102) FIG. **17**B illustrates a state in which a plurality of the carts **750** accommodated in line in the crossing direction (Y-direction are seen from above (Z-negative direction). As described above, the cart base portion **755** has the notch **755***a* and the protruding portion **755***c* on the one end side in the crossing direction and the notch **755***b* and the protruding portion **755***d* on the other end side. When the first cart **750**A and the second cart **750**B are accommodated by being aligned in the crossing direction, they can be disposed such that the protruding portion **755***d* of the first cart **750**A is fitted in the notch **755***a* of the second cart **750**B. At this time, they are disposed such that the protruding portion **755***c* of the second cart **750**B is fitted in the notch **755***b* of the first cart **750**A. As described above, by disposing the first cart **750**A and the second cart **750**B so that the cart base portions **755** of each of them overlap each other when seen from the width direction, the accommodation space can be reduced for a portion of a length Y1 in the crossing direction of the overlapping part. Moreover, since the cart base portion **755** of the first cart **750**A is fitted in the notch **755***a* of the cart base portion **755** of the second cart **750**B, the first cart **750**A and the second cart **750**B can be transported stably at the same time in the fitted state.

(103) By configuring such that the notch portions and the protruding portions of the plurality of carts are fitted as above, the accommodation space when the cart is not in use can be reduced as compared with the first embodiment. Therefore, according to this embodiment, more carts can be prepared easily and thus, the workability of mounting/removing of the roll is improved.

Eighth Embodiment

(104) Subsequently, an eighth embodiment will be described by referring to FIGS. **18**A and **18**B. The eighth embodiment is a configuration in which the accommodation space when the cart is not in use is reduced and is a mode different from the seventh embodiment. Note that a difference from the seventh embodiment is mainly described in the following, while the recording device **100** and other identical configurations are given the same signs, and the description thereof is omitted.

(105) FIG. **18**A is a perspective view of a state in which an extension/contraction mechanism **69** of

a cart **850** according to an eighth embodiment is fully extended. FIG. **18**B is a perspective view of a state in which the extension/contraction mechanism **69** of the cart **850** is contracted. By means of the cart **850**, as compared with the cart **750** according to the seventh embodiment, the carts can be accommodated with a further reduced grounding area when the cart is not in use. In the cart **850**, a receiving portion **851** and a cart base portion **855** are divided in the width direction (X-direction). On the divided cart base portions **855**, on each of which a protruding portion and a notch portion are formed, the receiving portion **851** and a height adjustment portion **853** are provided. Moreover, between the divided cart base portions **855**, the extension/contraction mechanism **69** capable of extension contraction is provided. That is, the cart **850** is configured such that a length in the width direction is changeable by extension/contraction of the extension contraction mechanism **69**.
(106) By configuring as above, as compared with the seventh embodiment, the accommodation space can be further reduced when the cart is not in use. Moreover, the cart in this embodiment has a configuration capable of handling a wider range of the roll widths as compared with the first embodiment and the seventh embodiment, since the width-direction length is changeable. Therefore, according to this embodiment, in addition to the reduction of the accommodation space when the cart is not in use, the range of the roll widths that can be handled can be widened.
(107) Note that application of the present invention is not limited to the aforementioned embodiments. For example, the configuration of each embodiment may be freely combined such that the receiving-portion roller **61** illustrated in the fourth embodiment is provided in the receiving portion **551** illustrated in the fifth embodiment.
(108) Examples of the configurations or concepts disclosed in the aforementioned embodiments are shown below. However, these are only exemplifications, and the aforementioned disclosures of the embodiments are not limited to the configurations or concepts shown below.
(109) Configuration A1
(110) A recording device in which a roll around which a sheet is wound is attached in a state supported by a cart, and an image is recorded on the sheet supplied from the roll, comprising:
(111) a roll support portion which supports the roll on an end part in a width direction of the roll; and
(112) a guide portion which extends in the width direction and supports the roll support portion movably in the width direction, wherein
(113) in a state in which the cart makes the roll movable in the width direction, a positioning portion which determines a position in a crossing direction crossing the width direction of a positioned portion provided in the cart with respect to the roll support portion is further included.
(114) Configuration A2
(115) The recording device described in the configuration A1, wherein
(116) the positioning portion is a surface which is orthogonal to the crossing direction, extends in the width direction and is in contact with the positioned portion.
(117) Configuration A3
(118) The recording device described in the configuration A1, wherein
(119) the positioning portion is a rail which extends in the width direction and is engaged with the positioned portion.
(120) Configuration A4
(121) A cart which supports a roll when the roll around which a sheet is wound is attached to a recording device, comprising:
(122) a roll receiving portion which supports the roll from below;
(123) a positioned portion whose position in a crossing direction crossing a width direction of the roll with respect to the recording device is determined by a positioning portion provided in the recording device; and
(124) a position adjustment mechanism capable of adjusting a position in the width direction of the roll supported by the roll receiving portion in a state in which the positioned portion is positioned

by the positioning portion.
(125) Configuration A5
(126) The cart described in the configuration A4, wherein
(127) the recording device further includes a height adjustment portion capable of adjusting a height of the roll receiving portion.
(128) Configuration A6
(129) The cart described in the configuration A4 or A5, wherein
(130) the positioned portion includes an abutting portion in contact with the positioning portion in the crossing direction.
(131) Configuration A7
(132) The cart described in the configuration A4 or A5, wherein
(133) the positioned portion is a lever mechanism capable of changing by swinging to a first attitude of being engaged with the positioned portion and a second attitude in which the engagement with the positioned portion is disengaged.
(134) Configuration A8
(135) The cart described in any one of the configuration A4 to A7, wherein
(136) the roll receiving portion includes two inclined surfaces forming a V-shape when viewed from the width direction; and
(137) the positioned portion is provided on an end part in the crossing direction of the cart.
(138) Configuration A9
(139) The cart described in any one of the configuration A4 to A8, wherein
(140) the roll receiving portion includes two inclined surfaces forming a V-shape when viewed from the crossing direction; and
(141) the positioned portion is further provided on an end part in the width direction of the cart.
(142) Configuration A10
(143) The cart described in the configuration A9, wherein
(144) two inclined surfaces forming a V-shape when viewed from the crossing direction are formed on end parts in the width direction of the roll receiving portion.
(145) Configuration A11
(146) The cart described in any one of the configuration A4 to A8, wherein
(147) the roll receiving portion is formed by being divided in the width direction;
(148) in the divided roll receiving portion, at least one is rotatable around a rotation axis extending in a vertical direction; and
(149) the positioned portion is provided on an end part in the crossing direction and on an end part in the width direction of the cart.
(150) Configuration A12
(151) The cart described in any one of the configuration A4 to A11, wherein
(152) the cart further includes a cart base portion which supports the roll receiving portion; and
(153) the position adjustment mechanism is a wheel portion provided on a lower part of the cart base portion.
(154) Configuration A13
(155) The cart described in the configuration A12, wherein
(156) the roll receiving portion includes a roller in contact with the roll and rotatable in the width direction.
(157) Configuration A14
(158) The cart described in any one of the configuration A4 to A11, wherein
(159) the cart further includes a cart base portion which supports the roll receiving portion; and
(160) the position adjustment mechanism is a slide mechanism capable of adjusting a position in the width direction of the roll receiving portion with respect to the cart base portion.
(161) Configuration A15

(162) The cart described in any one of the configuration A12 to A14, wherein
(163) the cart includes a protruding portion which protrudes downward from the cart base portion and has two inclined surfaces forming a V-shape at a distal end; and
(164) the protruding portion overlaps the roll receiving portion when viewed from a vertical direction.
(165) Configuration A16
(166) The cart described in any one of the configurations A12 to A15, wherein
(167) the cart base portion has a protruding portion which protrudes to one end side in the crossing direction and a notch portion formed on the other end side opposite to the one end side in the crossing direction,
(168) the protruding portion and the notch portion overlap each other when seen from the crossing direction, and
(169) in the width direction, a length of the notch portion is longer than a length of the protruding portion.
(170) Configuration A17
(171) The cart described in any one of the configurations A12 to A16, wherein the cart base portion is formed by being divided in the width direction, and
(172) the cart further includes an extension/contraction mechanism provided between the divided cart base portions and capable of extension/contraction in the width direction.
(173) Configuration A18
(174) A roll attachment mechanism which attaches a roll supported by a cart to a recording device which records an image on a sheet supplied from the roll around which the sheet is wound, wherein
(175) the recording device includes a roll support portion which supports the roll around which the sheet is wound on an end part in a width direction of the roll, a guide portion which extends in the width direction and movably supports the roll support portion in the width direction, and a positioning portion which determines a position of the cart with respect to the recording device in a crossing direction which extends in the width direction and crosses the width direction, and
(176) the cart includes a roll receiving portion which supports the roll from below, a positioned portion engaged with the positioning portion, and a position adjustment mechanism capable of adjusting a position in the width direction of the roll supported by the roll receiving portion in a state in which the positioning portion is engaged with the positioned portion.
(177) When the roll is to be conveyed to the sheet supply device prior to the setting of the roll, since it is difficult to carry a roll particularly with a high weight directly by hands, the roll is placed on the cart capable of supporting the roll from below for conveyance in many cases. At this time, if the roll can be set on the holder while it is still on the cart, there is no need to unload the roll from the cart even once from conveyance to the setting of the roll and workability of the roll attachment is improved. Thus, the sheet supply device and the recording device which can improve workability of the roll attachment to the sheet supply device will be described.
Ninth Embodiment
(178) Recording Device
(179) Frist, a basic configuration of a recording device **1100** including the sheet supply device according to a ninth embodiment of the present invention will be described by referring to FIGS. **19** and **20**. As the recording device, an inkjet recording device including a sheet supply device which supplies a sheet as a print medium and a recording portion which performs a recording operation such as print of an image on the sheet will be described as an application example. Moreover, a term “ink” in this description is used as a collective name of liquids such as recording liquids. Furthermore, when the same figure of various devices and the like which will be described below has the same or corresponding members in plural, suffixes such as a, b and the like are given and indicated in the figure but if there is no need to discriminate them in the explanation, the suffixes such as a, h and the like are omitted in the description.

(180) FIG. **19** is a front view of the recording device **1100**. The recording device **1100** includes a roll set portion **1200** which can send out the roll around which a sheet is wound in a roll state while supporting it as the sheet supply device. In the recording device **1100**, a roll set portion **1200***a* on the paper-feed side and a roll set portion **1200***b* on the take-up side are provided. An image can be printed on the sheet supplied from the roll supported by the roll set portion **1200***a*, and the roll set portion **1200***b* can take up the printed sheet. A user can input various commands and the like such as specification of a size of the roll, setting of a type of the roll and the like to the recording device **1100** by using the various switches provided on an operation panel **1013**. Note that, as shown in FIG. **19**, a width direction of the roll supported by the recording device **1100** is defined as an X-axis direction, a depth direction of the device as a Y-axis direction, and a gravity direction as a Z-axis direction, respectively, in the following explanation. In this embodiment, the X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to one another.
(181) FIG. **20** is a schematic sectional view of the recording device **1100** illustrating a conveyance path of the sheet S. The sheet S pulled out of the roll R set on (attached to) the roll set portion **1200***a* is connected to the roll set portion **1200***b* and is taken up. The roll set portion **1200** includes a roll holder **1002** and a holder guide **1003**. The roll set portion **1200**, the roll holder **1002**, and the holder guide **1003** provided on the paper-feed side and the take-up side, respectively, will be described by discriminating them as necessary by giving "a" to the end of a sign of a member on the paper-feed side and "b" to the end of a sign of a member on the take-up side.
(182) The sheet S pulled out of the roll R and set on the roll set portion **1200***a* is conveyed by an upstream sheet conveying portion **1300** to a print portion **1400** as a recording portion capable of printing an image. In the print portion **1400**, the image is printed on the sheet S by ejection of the ink from an inkjet-type printhead **1008**.
(183) The printhead **1008** ejects the ink from an ejection port by using an ejection energy-generating element such as an electricity-heat conversion element (heater), a piezo element and the like. When the electricity-heat conversion element is used, the ink is foamed by heat generation thereof, and the ink can be ejected from the ejection port by using the foaming energy. In an application of the present invention, the printhead **1008** is not limited only to an inkjet type. Moreover, a print type of the print portion **1400** is not limited, either, and it may be a serial scan type, a full-line type and the like, for example. In the case of the serial scan type, the image is printed with a conveying operation of the sheet S and scanning of the printhead **1008** in a direction crossing the conveying direction of the sheet S. In the case of the full-line type, the printhead **1008** which is lengthy and extends in the direction crossing the conveying direction of the sheet S is used, and an image is printed while the sheet S is continuously conveyed.
(184) In a hollow hole portion of the roll R, a shaft-shaped paper-core fixing portion **1011** provided on the roll holder **1002** is inserted. The paper-core fixing portion **1011** is rotated forward and reversely and driven by transmission of power from a roll drive motor, not shown, provided on the roll holder **1002**. The roll is held at a center part by the paper-core fixing portion **1011** and is rotated in arrow C**1**, C**2** directions. By enabling driving in both directions of forward rotation and reverse rotation, print can be performed selectively on a front surface or a back surface of the sheet S as shown by a path indicated by a solid line and a dotted line continuing from the roll set portion **1200***a* on the paper-feed side in FIG. **20** to the upstream sheet conveying portion **1300**. Similarly, the printed surface of the sheet S can be selectively taken up on an outer side or an inner side as shown by a path indicated by a solid line and a dotted line continuing from a downstream sheet conveying portion **1500** in FIG. **20**, which will be described later, to the roll set portion **1200***b* on the take-up side.
(185) An upstream conveyance guide **1004** of the upstream sheet conveying portion **1300** leads the sheet S to the print portion **1400** while guiding the front surface and the back surface of the sheet S pulled out of the roll set portion **1200***a*. A conveyance roller **1005** is rotated forward and reversely by a conveyance-roller drive motor. A nip roller **1006** is capable of being driven/rotated in

accordance with rotation of the conveyance roller **1005**. A conveying speed of the sheet S by the conveyance roller **1005** is set higher than a pulling-out speed of the sheet S by the rotation of the roll R, whereby a back tension is given to the sheet S so that the sheet S can be conveyed in a state in which a tension is applied to the sheet S. As a result, sagging of the sheet S is prevented, and occurrence of a crease in the sheet S and occurrence of a conveyance error can be suppressed.
(186) A platen **1007** in the print portion **1400** adsorbs a back surface on a side opposite to a recording surface of the sheet S through a suction hole **1007***a* provided in the platen **1007** by a negative pressure generated by a suction fan **1009**. As a result, by regulating the position of the sheet S so that it goes on the platen **1007**, high-precision print of an image by the printhead **1008** can be made possible.
(187) A downstream conveyance guide **1010** of the downstream sheet conveying portion **1500** leads the sheet S to the roll set portion **1200***b* on the take-up side while guiding the front surface and the back surface of the sheet S pulled out of the print portion **1400**. Here, a distal end of the sheet S is fixed to a paper core which was set on the roll set portion **1200***b*, and by rotating the paper core set in accordance with the conveying speed of the conveyance roller **1005**, the sheet S printed by the print portion **1400** can be continuously taken up.
(188) The recording device **1100** includes a stand **1014** on both end parts in the width direction, respectively. The stand **1014** supports weights of the upstream sheet conveying portion **1300**, the print portion **1400**, and the downstream sheet conveying portion **1500** on an upper part of a main body from below and supports the weight of the roll R through the roll set portion **1200**. The stand **1014** is connected to the holder guide **1003** of the roll set portion **1200** and also functions as a guide support portion which supports the holder guide **1003**. The stand **1014** is in contact with a floor surface and releases the weight of the recording device **1100** to the floor surface.
(189) Roll Set Portion
(190) Subsequently, by referring to FIGS. **21**A to **21**C, and **22**, details of the roll set portion **1200** according to this embodiment will be described. FIGS. **21**A to **21**C are explanatory diagrams illustrating a state in which the roll R is set on the roll set portion **1200**. FIG. **21**A illustrates a state in which the paper-core fixing portion **1011** of a reference roll holder **1021** is inserted into one end in the width direction of the roll R. FIG. **21**B illustrates a state in which the paper-core fixing portion **1011** of a non-reference roll holder **1022** is inserted into the other end of the roll R whose one end is supported. FIG. **21**C illustrates a state in which the non-reference roll holder **1022** is fixed by a lock mechanism **1012**, and support of the roll R is completed.
(191) The roll holder **1002** provided on the roll set portion **1200** is constituted by a reference roll holder **1021** and a non-reference roll holder **1022** which make a pair. The reference roll holder **1021** supports the one end side in the width direction of the roll R, while the non-reference roll holder **1022** supports the other end side in the width direction of the roll R so that the roll R is supported by the roll holder **1002**. The paper-core fixing portion **1011** is rotatably mounted on each of the reference roll holder **1021** and the non-reference roll holder **1022**. The non-reference roll holder **1022** is held by the holder guide **1003** as the guide portion slidably in the width direction (X-direction) of the roll R. Moreover, the lock mechanism **1012** is provided on the non-reference roll holder **1022**, and by turning a knob provided on the lock mechanism **1012** in a T direction, the non-reference roll holder **1022** can be fixed at an arbitrary position in the width direction with respect to the holder guide **1003**. The reference roll holder **1021** is also held by the holder guide **1003**, but the slide mechanism is not indispensable. A friction member is provided in the paper-core fixing portion **1011**, and when the roll R is set, the roll R and the paper-core fixing portion **1011** are rotated integrally. In the reference roll holder **1021**, a roll drive motor, not shown, is provided in connection to the paper-core fixing portion **1011**. By transmission of a rotary force by the roll drive motor to the paper-core fixing portion **1011**, the paper-core fixing portion **1011** on the reference side is rotated forward and reversely and driven. The paper-core fixing portion **1011** on the non-reference side does not have the drive motor connected and is driven and rotated by the drive

transmitted through the paper core of the roll R, when the roll R is set.
(192) When the roll R is to be set on the roll set portion **1200**, first, as shown in FIG. **21**A the paper-core fixing portion **1011** of the reference roll holder **1021** is inserted into one of the paper cores of the roll R. When the one of the paper cores of the roll R is brought into contact with the paper-core fixing portion **1011**, the friction member of the paper-core fixing portion **1011** is fitted in the paper core of the roll R. As a result, the roll R is fixed to the paper-core fixing portion **1011** of the reference roll holder **1021**. Subsequently, as shown in FIG. **21**B, the non-reference roll holder **1022** is slid in the width direction, and the paper-core fixing portion **1011** of the non-reference roll holder **1022** is inserted into the other paper core of the roll R. As a result, similarly to the reference side, the roll R is fixed to the paper-core fixing portion **1011** of the non-reference roll holder **1022**. Lastly, as shown in FIG. **21**C, the non-reference roll holder **1022** is fixed to the holder guide **1003** by using the lock mechanism **1012**. As a result, sliding of the non-reference roll holder **1022** in the paper width direction, which causes removal of the roll R, is prevented.
(193) By holding the roll R by using the slide mechanism by the roll holder **1002** and the holder guide **1003**, the roll R is firmly supported by the roll set portion **1200** regardless of the roll width. That is, the roll set portion **1200** is configured to be capable of setting the roll R with an arbitrary width not larger than a length in the width direction of the holder guide **1003**. Moreover, in the case of a configuration in which the reference roll holder **1021** is fixed to the holder guide **1003** and is not slid, since the end part of the roll R is fitted in the fixed reference roll holder **1021**, the reference-side end part position of the roll R is configured to be determined constant with respect to the roll set portion **1200**.
(194) The setting of the roll R to the roll set portion **1200** shown in FIGS. **21**A to **21**C has been described above, and by performing this setting procedure in a reverse order, the roll R can be removed from the roll set portion **1200**.
(195) FIG. **22** is a perspective view of the roll set portion **1200** according to the ninth embodiment. To the stand **1014**, a holder guide **1003***a* on the paper-feed side and a holder guide **1003***b* on the take-up side are fixed. The roll holder **1002***a* on the paper-feed side is configured such that a reference roll holder **1021***a* (first roll support portion) and a non-reference roll holder **1022***a* (second roll support portion) are disposed on a front side of the recording device (Y-positive direction side) with respect to the holder guide **1003***a*. That is, the holder guide **1003***a* is disposed on a main-body depth side (Y-negative direction side) with respect to the paper-core fixing portion **1011** on the paper-feed side. By having such configuration, since there is no structure such as the holder guide **1003** or the like on a main-body front side or a main-body lower side of the paper-core fixing portion **1011**, the roll holder **1002***a* on the paper-teed side does not make an obstacle in the mounting work of the roll R. Therefore, in the recording device **1100**, movement in an entry direction P shown in FIG. **22** with the roll R placed on the cart and setting with the roll R still on the cart are made possible. In this embodiment, the entry direction P is parallel to the depth direction (Y-direction), and an upstream side in the entry direction P is the device front side, and a downstream side is a device depth side.
(196) Moreover, the roll holder **1002***b* on the take-up side is configured such that a reference roll holder **1021***b* (third roll support portion) and a non-reference roll holder **1022***b* (fourth roll support portion) are disposed on the main-body depth side with respect to the holder guide **1003***b*. That is, in the recording device **1100**, four roll support portions are provided. Regarding the roll holder **1002***b* on the take-up side, since there is no structure such as the holder guide **1003** or the like on the main-body depth side or the main-body lower side of the paper-core fixing portion **1011**, workability of the removal of the roll R is favorable.
(197) Cart
(198) FIG. **23** is a perspective view of a cart **1050** according to the ninth embodiment. The cart **1050** includes a receiving portion **1051** which can support the roll R from below in a state in which the width direction of the roll R is horizontal, a wheel portion **1052** for movement, and a height

adjustment portion **1053** which can adjust a height position of the receiving portion **1051**. A length in the width direction (X-direction) of the receiving portion **1051** is configured to be smaller than the roll width of the roll R to be set so that the roll holder **1002** does not interfere with the receiving portion **1051** at the roll setting. Moreover, on the receiving portion **1051**, an inclined surface in order to prevent rolling and falling of the roll R is provided. There is no particular limitation on a detailed form of the cart **1050**, but it is preferable that the height of the receiving portion **1051** can be adjusted by the height adjustment portion **1053** or the like so that various roll diameters can be handled.

(199) FIGS. **24**A to **24**D are front views illustrating a procedure of setting the roll R on the roll set portion **1200** by using the cart **1050**. FIG. **24**A illustrates a state in which, after the cart **1050** is moved to a space between the opposing paper-core fixing portions **1011**, the roll R is positioned in the vertical direction. FIG. **241**B illustrates a state in which the cart **1050** on which the roll R is placed is moved to the reference roll holder **1021** side. FIG. **24**C illustrates a state in which the non-reference roll holder **1022** is moved to the roll R side in a state in which the one end in the width direction of the roll R is in contact with the paper-core fixing portion **1011**. FIG. **24**D illustrates a state in which the non-reference roll holder **1022** is fixed by the lock mechanism **1012**, and the support of the roll R is completed.

(200) When the roll R placed on the receiving portion **1051** of the cart **1050** is to be set on the roll set portion **1200**, first, the cart **1050** is moved so that the roll R is positioned in a region sandwiched by the two roll holders **1002**. After the position of the roll R in the depth direction (Y-direction) is determined, as in FIG. **24**A, the height of the center of the roll R is matched with the height of the paper-core fixing portion **1011** of the recording device **1100**, and the positioning in the vertical direction is performed. The height adjustment portion **1053** is provided on the cart **1050**, and the height (Z-direction position) of the roll R is adjusted by using it. Subsequently, as in FIG. **24**B, by moving the cart **1050** in the width direction, the one end of the paper core of the roll R is inserted into the paper-core fixing portion **1011** of the reference roll holder **1021**. Since both of a depth-direction position and a vertical-direction position of the center of the roll R match each other with respect to the paper-core fixing portion **1011** by the operations so far, the paper core can be easily inserted into the paper-core fixing portion **1011**. When the one end of the paper core of the roll R and the paper-core fixing portion **1011** contact each other, the friction member of the paper-core fixing portion **1011** is fitted in the paper core of the roll R. As a result, the roll R is fixed to the paper-core fixing portion **1011** of the reference roll holder **1021**. Moreover, as in FIG. **24**C, the non-reference roll holder **1022** is slid in the width direction so as to get closer to the roll R, and the other end of the paper core of the roll R is inserted into the paper-core fixing portion **1011** of the non-reference roll holder **1022**. As a result, similarly to the reference roll holder **1021**, the roll R is fixed to the paper-core fixing portion **1011** of the non-reference roll holder **1022**, and the width-direction (X-direction) position of the roll R is determined. Lastly, as in FIG. **24**D, the non-reference roll holder **1022** is fixed to the holder guide **1003** by using the lock mechanism **1012**. As a result, sliding of the non-reference roll holder **1022** in the width direction and removal of the roll R from the roll holder **1002** are prevented. Subsequently, the receiving portion **1051** is lowered downward by the height adjustment portion **1053**, and the cart **1050** is retreated from the roll R so that the setting of the roll R on the recording device **1100** is completed.

(201) The setting procedure of the roll R on the roll set portion **1200** by using the cart **1050** has been described in FIGS. **24**A to **24**D so far, and by performing this setting procedure in a reverse order, the roll R can be removed from the roll set portion **1200** without manually supporting the roll R. By configuring such that the cart **1050** can be used for the removal of the roll R, operations from the removal of the roll R from the recording device **1100** to the conveyance of the roll R can be performed without unloading the roll R from the cart **1050**. Moreover, by setting the roll, which has one side printed and was removed from the roll set portion **1200***b* on the take-up side by using the cart **1050**, on the roll set portion **1200***a* on the paper-feed side while the roll remains on the cart

1050, double-sided printing can be performed without unloading the roll R from the cart **1050**.
(202) FIGS. **25**A and **25**B are diagrams illustrating a space in Which the roll R is set or a space into which the cart **1050** is made to enter in the recording device **1100**. In a space between the reference roll holder **1021** and the non-reference roll holder **1022** of the roll holder **1002** oppose each other, a first space D**1** in which the roll R is installed is formed. Below the first space D**1**, there is no obstacle when the cart **1050** is to be disposed, and a second space D**2** into which the cart **1050** enters at the roll set is formed. The cart **1050** on which the roll R is placed moves from the device front side to the device depth side and enters the second space D**2**. Note that each of the spaces described above does not mean that there is no object at all but it is only necessary that obstacles or the like are not present to such a degree that the respective objects can be achieved.
(203) As described above, according to this embodiment, since the roll R can be set with the roll R placed on the cart **1050**, the operations from the conveyance of the roll R to the setting thereof on the roll set portion **1200** can be performed without unloading the roll R from the cart **1050**. Moreover, when the roll is to be set, since the weight of the roll R is supported by the cart **1050** at all times, the roll R can be set without manually supporting the weight of the roll R. That is, the workability of the roll attachment to the roll holder **1002** of the roll set portion **1200** is improved.
Tenth Embodiment
(204) Subsequently, a tenth embodiment will be described by referring to FIG. **26**. In the tenth embodiment, the positional relationship between the roll set portion **1200***a* on the paper-teed side and the roll set portion **1200***b* on the take-up side is different from that of the ninth embodiment, Note that a difference from the ninth embodiment is mainly described in the following, while the other identical configurations are given the same signs, and the description thereof is omitted.
(205) FIG. **26** is a side view of a recording device **1600** according to the tenth embodiment. In FIG. **26**, an operation from removal to resetting of the roll R by using the cart **1050** is illustrated. Such operation is performed when the roll R in which one side has been printed in double-sided printing is switched from the roll set portion **1200***b* to the roll set portion **1200***a*. In the tenth embodiment, the roll set portion **1200***a*, the roll set portion **1200***b*, and the paper-core fixing portions **1011** provided on each of them are configured with the same height. That is, the rotation center of the roll supported by the roll set portion **1200***a* and the rotation center of the roll supported by the roll set portion **1200***b* match each other in height.
(206) In the case of the double-sided print, after the print is performed on one side of the sheet, the taken-up roll needs to be set on the roll set portion on the paper-feed side again. At this time, in this embodiment, the roll R is removed from the roll set portion **1200***b* on the take-up side and is placed on the cart **1050**, the cart **1050** with the roll R placed thereon is moved to the roll set portion **1200***a* on the paper-feed side, and the roll R can be set on the roll set portion **1200***a*. That is, after the one-side print, without manually lifting up the roll R and without adjusting the height of the cart **1050**, the work from the removal to the mounting of the roll R can be performed. If the heights of the paper-core fixing portions **1011** of the roll set portion **1200***a* and the roll set portion **1200***b* are different, when the roll R is to be re-set on the roll set portion **1200***a*, the height of the cart **1050** needs to be adjusted or the roll R needs to be lifted up temporarily. Thus, by setting the heights of the paper-core fixing portions **1011** of the roll set portion **1200***a* and the roll set portion **1200***b* the same as in this embodiment, the series of works of the removal, conveyance, and re-setting of the roll R can be performed without adjusting the height of the roll R and thus, the workability is improved.
Eleventh Embodiment
(207) Subsequently, an eleventh embodiment will be described by referring to FIGS. **27**A and **27**B. In the eleventh embodiment, a sliding configuration of the non-reference roll holder **1022** is different from that of the ninth embodiment. Note that a difference from the ninth embodiment is mainly described in the following, while the other identical configurations are given the same signs, and the description thereof is omitted.

(208) FIG. **27**A is a sectional view illustrating a slide mechanism of the non-reference roll holder **1022**. The non-reference roll holder **1022** on which the paper-core fixing portion **1011** is provided is capable f sliding in the width direction (X-direction) along the holder guide **1003**. The non-reference roll holder **1022** in this embodiment has a plurality of holder rollers **1031** as roller portions in contact with the holder guide **1003**. The non-reference roll holder **1022** is supported by the holder guide **1003** through the holder rollers **1031**. The holder rollers **1031** are fixed rotatably around a rotation axis extending in a direction orthogonal to the width direction with respect to the non-reference roll holder **1022**. By means of rotation of the holder rollers **1031** connected to the non-reference roll holder **1022** and in contact with the holder guide **1003**, sliding resistance of the non-reference roll holder **1022** at the sliding is reduced.
(209) In the non-reference roll holder **1022**, a holder roller **1031***a* (first roller), a holder roller **1031***b* (second roller), and a holder roller **1031***c* (third roller) are provided. The holder roller **1031***a* is brought into contact with the holder guide **1003** downward (Z-positive direction) from the upper side in the vertical direction. The holder roller **1031***b* is brought into contact with an upper end part of the holder guide **1003** by being directed to the front (Y-positive direction) from a depth side in the depth direction. The holder roller **1031***c* is brought into contact with a lower end part of the holder guide **1003** below the holder roller **1031***b* by being directed to the depth (Y-negative direction) from the front side in the depth direction. Moreover, the holder roller **1031***a*, the holder roller **10311**), the holder roller **1031***c* are provided in plural rows in the width direction. By disposing the holder rollers **1031** as above, the non-reference roll holder **1022** is supported not to be inclined with respect to a moment generated by the weight of the non-reference roll holder **1022** located closer to the main-body front side than the holder guide **1003**. Moreover, by disposing one row or more of the holder rollers **1031** in each of an outer side and an inner side with respect to the contact part between the paper-core fixing portion **1011** and the paper core in the width direction, the non-reference roll holder **1022** can slide in the width direction without nibbling the holder guide **1003**.
(210) FIG. **27**B is a sectional view illustrating the slide mechanism of the non-reference roll holder **1022** of an example different from the example shown in FIG. **27**A in the eleventh embodiment. In the example shown in FIG. **27**A, a sectional shape of the holder guide **1003** is a rectangle, but in the example shown in FIG. **27**B, two pieces of holder guides **1032** having a circular sectional shape are provided by being aligned in the vertical direction. In this configuration, too, by configuring a contact direction and a contact position of the holder roller **1031** with respect to the holder guide **1032** similarly to the configuration described above, the sliding resistance at the sliding can be reduced while the non-reference roll holder **1022** is slidably supported in an appropriate attitude.
(211) Note that the shapes of the roll holder and the holder guide, the number and disposition configuration of the holder rollers are not limited to the configuration described above but are capable of various changes. For example, the number of holder rollers may be increased for further stabler support of the roll holders.
Twelfth Embodiment
(212) Subsequently, a twelfth embodiment will be described by referring to FIGS. **28**, **29**, **30**A, and **30**B. The twelfth embodiment is different from the ninth embodiment in a point that a position adjustment mechanism that adjusts the position of the roll holder **1002** for alignment adjustment is provided. Note that a difference from the ninth embodiment is mainly described in the following, while the other identical configurations are given the same signs, and the description thereof is omitted.
(213) FIG. **28** is a side view of a recording device **1700** according to the twelfth embodiment. FIG. **29** is a perspective view of a roll set portion **1750** according to the twelfth embodiment. In order to prevent skewing of the sheet S, the alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** is needed. However, in the configurations of the embodiments described so far, the conveyance roller **1005** and the paper-core fixing portion **1011** are fixed

through a plurality of components, and when the conveyed sheet S is inclined, the alignment adjustment is difficult. Thus, in this embodiment, a position of a guide adjustment portion **1015**, which is a connection member which connects the stand **1014** to the holder guide **1003**, with respect to the stand **1014**, is configured to be capable of being adjusted. By adjusting the position of the guide adjustment portion **1015**, a fixed position in the depth direction (Y-direction) and in the vertical direction (Z-direction) of the holder guide **1003** with respect to the stand **1014** can be adjusted. Therefore, by adjusting the position of the guide adjustment portion **1015** with respect to the holder guide **1003**, the alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** is made, whereby the skewing of the sheet S can be prevented. In the recording device **1700**, a guide adjustment portion **1015***a* connected to the holder guide **1003***a* on the paper-feed side and a guide adjustment portion **1015***b* connected to the holder guide **1003***b* on the take-up side are provided.

(214) FIG. **30**A is a perspective view illustrating the guide adjustment portion **1015***a*, and the stand **1014** in a separated manner, and FIG. **30**B is a perspective view illustrating a state in which the guide adjustment portion **1015***a* is fixed to the stand **1014**. The position adjustment mechanism in this embodiment includes the guide adjustment portion **1015** in which a mounting hole **1151** is provided and the stand **1014** in which a screw hole **1141** is provided. When a bolt **1061** is inserted into the mounting hole **1151** and the screw hole **1141**, and they are tightened by the bolt **1061** through a washer **1062**, the guide adjustment portion **1015** is fixed to the stand **1014**. The mounting hole **1151** is formed larger than the screw portion of the bolt **1061**, and by slightly adjusting the position in the depth direction and the vertical direction of the guide adjustment portion **1015** with respect to the stand **1014**, the guide adjustment portion **1015** can be fixed to the stand **1014**. That is, if the conveyed sheet S is skewed, the alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** can be made by adjusting the position of the guide adjustment portion **1015**. Note that, if a diameter of a head part of the bolt **1061** is sufficiently larger than the mounting hole **1151**, and the guide adjustment portion **1015** can be sufficiently fixed, the washer **1062** does not have to be provided.

(215) In this embodiment, the guide adjustment portion **1015** is provided only on the end part on the non-reference roll holder **1022** side of the holder guide **1003**, but it may be configured to be provided on the end part on the reference roll holder **1021** side. That is, as the alignment adjustment mechanism, the position adjustment mechanism such as the guide adjustment portion or the like needs to be provided at least on one end part of the holder guide, but it may be configured to be provided on either one of the reference side and the non-reference side or on both ends on the reference side and the non-reference side.

Thirteenth Embodiment

(216) Subsequently, a thirteenth embodiment will be described by referring to FIGS. **31** to **33**. In the thirteenth embodiment, a position adjustment mechanism with a configuration different from that of the twelfth embodiment is provided, and alignment can be adjusted. Note that a difference from the ninth embodiment is mainly described in the following, while the other identical configurations are given the same signs, and the description thereof is omitted.

(217) FIG. **31** is a side view of a recording device **1800** according to the thirteenth embodiment. FIG. **32** is a perspective view of a roll set portion **1850** according to the thirteenth embodiment. In a configuration in which the non-reference roll holder **1022** is movable in the width direction, it is preferable that alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** can be made regardless of the position in the width direction of the non-reference roll holder **1022**. Thus, the roll set portion **1850** of this embodiment is configured such that, as the position adjustment mechanism of the roll holder **1002**, a plurality of holder sliders **1016** extending in substantially the whole region in the width direction of the holder guide **1003** are provided. That is, the recording device **1800** is configured to be capable of the alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** substantially the whole region in the

width direction of the holder guide **1003** by the holder sliders **1016**.

(218) The holder sliders **1016** are provided three each on the respective holder guides **1003** on the paper-feed side and the take-up side and are slide members for the non-reference roll holder **1022***a* on the paper-feed side and the non-reference roll holder **1022***b* on the take-up side to slide in the width direction. The holder guide **1003***a* on the paper-feed side supports the roll holder **1002***a* through the holder slider **1016***a* (first slider), a holder slider **1016***b* (second slider), and a holder slider **1016***c* (third slider). The holder guide **1003***b* on the take-up side supports the roll holder **1002***b* through a holder slider **1016***d*, a holder slider **1016***e*, a holder slider **1016***f*.

(219) The holder slider **1016***a* is provided on an upper end of the holder guide **1003***a* and is in contact with the roll holder **1002***a* upward in the vertical direction (Z-negative direction). The holder slider **1016***b* is provided above and on a depth side in the depth direction of the holder guide **1003***a* and is in contact with the roll holder **1002***a* toward the depth in the depth direction (Y-negative direction). The holder slider **1016***c* is provided below and on a front side in the depth direction of the holder guide **1003***a* and is in contact with the roll holder **1002***a* toward the front in the depth direction (Y-positive direction). By disposing the holder sliders **1016** as above, the roll holder **1002** can be supported so as not to be inclined. Moreover, the depth-direction (Y-direction) position of the roll holder **1002** can be adjusted by the holder slider **1016***b* and the holder slider **1016***c*. Furthermore, the vertical-direction (Z-direction) position of the roll holder **1002** can be adjusted by the holder slider **1016***a*. As described above, by adjusting the position and the attitude of the roll holder **1002** on which the paper-core fixing portion **1011** is provided by the holder sliders **1016**, the alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** can be made.

(220) Subsequently, details of the position adjustment method by the holder sliders **1016** will be described by using the holder slider **1016***a* as an example. FIG. **33** is a perspective view illustrating the adjustment method of the vertical-direction position of the roll holder **1002** by the holder slider **1016***a*. In FIG. **33**, the holder slider **1016***a* before the adjustment is indicated by a solid line, and the holder slider **1016***a* after the adjustment is indicated by a dotted line. The holder slider **1016***a* after the adjustment shown in FIG. **31** has been adjusted in a direction in which the vertical-direction position of the roll holder **1002** is adjusted downward.

(221) The holder slider **1016***a* is formed of an elongated plate material bent in the short-side direction and has a plurality of surfaces extending in the width direction. The holder slider **1016***a* has a first surface **1161** in contact with the roll holder **1002**, a second surface **1162** in parallel with the first surface **1161** and in contact with the holder guide **1003**, and a third surface **1163** orthogonal to the second surface **1162** and in contact with the holder guide **1003**. The second surface **1162** is in contact with the holder guide **1003** downward in the vertical direction (Z-positive direction), and the third surface **1163** is in contact with the holder guide **1003** toward the depth in the depth direction (Y-negative direction). The holder slider **1016***a* further has an adjustment long-hole **1164** which penetrates the second surface **1162** and is long in the width direction, a mounting long-hole **1165** which penetrates the second surface **1162** and is long in the vertical direction, and a mounting long-hole **1166** which penetrates the third surface **1163** and is long in the depth direction. By inserting a screw component such as a bolt into a screw hole (not shown) of the holder guide **1003** through the mounting long-hole **1165** and the mounting long-hole **1166**, the holder slider **1016***a* is fixed to the holder guide **1003**.

(222) For the position adjustment by the holder slider **1016***a*, an adjustment jig **1019** engaged with both a reference hole **1018** provided in the holder guide **1003** and the adjustment long-hole **1164** is used. The reference hole **1018** is located on an inner side of the adjustment long-hole **1164** when viewed from an axial direction of the reference hole **1018**. The adjustment jig **1019** has a cylindrical shaft portion **1191** and a pin portion **1192** provided at a distal end of the shaft portion **1191**. A center axis of the pin portion **1192** is disposed by being shifted with respect to a center axis of the shaft portion **1191**, and the pin portion **1192** is eccentric to the shaft portion **1191**. By

inserting the pin portion **1192** into the reference hole **1018**, by inserting the shaft portion **1191** into the adjustment long-hole **1164**, and by rotating the adjustment jig **1019** around the center axis of the pin portion **1192**, the adjustment long-hole **1164** is pressed onto the shaft portion **1191** in the vertical direction. That is, by adjusting the adjustment jig **1019** in a state in which the fixing by a screw component inserted into the mounting long-hole **1165** or the mounting long-hole **1166** is relaxed, the vertical-direction position of the first surface **1161** of the holder slider **1016** is adjusted.
(223) When the position of the first surface **1161** in contact with the roll holder **1002** is adjusted, the positions of the second surface **1162** and the third surface **1163** are also changed with respect to the holder guide **1003**. For example, when the position of the first surface **1161** is to be adjusted downward, the second surface **1162** is also moved downward, and the third surface moves to the front side in the depth direction. Even in such a case, the mounting long-hole **1165** provided in the second surface **1162** and the mounting long-hole **1166** provided in the third surface **1163** are formed having a long-hole shape so that the holder slider **1016** can be fixed to the holder guide **1003**.
(224) By providing a plurality of the reference holes **1018** in the width direction of the holder guide **1003** and by providing a plurality of the adjustment long-holes **1164**, the mounting long-holes **1165**, and the mounting long-holes **1166** in the width direction of the holder slider **1016***b*, the alignment adjustment in the whole region in the width direction can be made with high accuracy.
(225) The method of adjusting the position of the roll holder **1002** has been described above with the holder slider **1016***b* as an example, but the configurations and the position adjustment methods of the other holder sliders **1016** are also the same. By providing the plurality of holder sliders **1016** with different contact direction with respect to the roll holder **1002** on the holder guide **1003**, the alignment adjustment of the conveyance roller **1005** and the paper-core fixing portion **1011** can be made by adjusting the position and the attitude of the roll holder **1002**.
(226) Note that the application of the present invention is not limited to the embodiments described above. The configuration of each embodiment may be freely combined such as a configuration capable of reducing the sliding resistance at slide and of performing the alignment adjustment by providing the holder roller shown in the eleventh embodiment and the holder slider **1016** shown in the thirteenth embodiment, for example.
(227) Examples of configurations or concepts disclosed in this embodiment described above are shown below. However, these are only exemplification, and the configurations or concepts shown below by the disclosure of this embodiment described above are not limiting.
(228) Configuration B1
(229) A sheet supply device that rotates a roll around which a sheet is wound while supporting the same and sends out the sheet, comprising:
(230) a first roll support portion which rotatably supports the roll on one end side in a width direction of the roll;
(231) a second roll support portion which is provided to face the first roll support portion and rotatably supports the roll on the other end side of the roll; and
(232) a guide portion which extends in the width direction and movably supports the second roll support portion in the width direction, wherein
(233) below a first space in which the roll between the first roll support portion and the second roll support portion is installed, a second space into which a cart which transports the roll enters is formed.
(234) Configuration B2
(235) The sheet supply device described in the configuration B1, wherein
(236) the guide portion is located on a downstream side in an entry direction of the cart with respect to the first space.
(237) Configuration B3
(238) The sheet supply device described in the configuration B2, wherein

(239) the second roll support portion has a roller portion in contact with the guide portion and rotatable around a rotation axis extending in a direction orthogonal to the width direction.
(240) Configuration B4
(241) The sheet supply device described in the configuration B3, wherein
(242) the roller portion includes a first roller in contact with the guide portion from an upper side in a vertical direction, a second roller in contact with the guide portion from a downstream side in the entry direction, and a third roller in contact with the guide portion below the second roller from an upstream side in the entry direction.
(243) Configuration B5
(244) The sheet supply device described in any one of the configuration B2 to B4, further comprising:
(245) a guide support portion which supports the guide portion; and
(246) a position adjustment mechanism capable of adjusting a position in the entry direction and a vertical direction of at least either one of the first roll support portion and the second roll support portion with respect to the guide support portion.
(247) Configuration B6
(248) The sheet supply device described in the configuration B5, wherein
(249) the position adjustment mechanism includes a connection member which connects the guide portion and the guide support portion and can adjust the position in the entry direction and the vertical direction of at least either one of the first roll support portion and the second roll support portion with respect to the guide support portion.
(250) Configuration B7
(251) The sheet supply device described in the configuration B5, wherein
(252) the position adjustment mechanism includes a slide member which extends in the width direction, is provided on the guide portion, and can adjust the position in the entry direction and the vertical direction of at least either one of the first roll support portion and the second roll support portion with respect to the guide portion by being brought into contact with the second roll support portion.
(253) Configuration B8
(254) The sheet supply device described in the configuration B7, wherein
(255) the slide member includes a first slider provided on an upper end of the guide portion, a second slider provided on a downstream side in the entry direction with respect to the guide portion, and a third slider provided below the second slider and on an upstream side in the entry direction with respect to the guide portion.
(256) Configuration B9
(257) A recording device, comprising:
(258) a sheet supply device that rotates a roll around which a sheet is wound while supporting the same and sends out the sheet; and
(259) a recording portion which performs a recording operation on the sheet sent out of the sheet supply device, wherein
(260) the sheet supply device includes:
(261) a first roll support portion which rotatably supports the roll on one end side in a width direction of the roll;
(262) a second roll support portion which is provided to face the first roll support portion and rotatably supports the roll on the other end side of the roll; and
(263) a guide portion which extends in the width direction and movably supports the second roll support portion in the width direction, wherein
(264) below a first space in which the roll between the first roll support portion and the second roll support portion is installed, a second space into which a cart which transports the roll enters is formed.

(265) Configuration B10
(266) The recording device described in the configuration B9, wherein
(267) the recording device further includes:
(268) a third roll support portion which rotatably supports the roll on the one end side of the roll; and
(269) a fourth roll support portion which is provided to face the third roll support portion and rotatably supports the roll on the other end side of the roll, wherein
(270) the sheet for which the recording operation was performed is taken up by the roll installed between the third roll support portion and the fourth roll support portion.
(271) Configuration B11
(272) The recording device described in the configuration B10, wherein
(273) a rotation center of a roll supported by the first roll support portion and the second roll support portion matches a rotation center of a roll supported by the third roll support portion and the fourth roll support portion in height.
(274) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
(275) This application claims the benefit of Japanese Patent Application No. 2022-017460, filed on Feb. 7, 2022, and Japanese Patent Application No. 2022-029174, filed on Feb. 28, 2022, which are hereby incorporated by reference herein in their entirety.

Claims

1. A recording device, comprising: a recording portion configured to record images on a sheet supplied from a roll on which the sheet is wound; a first roll support portion which rotatably supports the roll at a front side of the recording device for supplying the sheet to the recording portion on one end side in a width direction of the roll; a second roll support portion which is provided to face the first roll support portion and rotatably supports the roll at the front side of the recording device for supplying the sheet to the recording portion on the other end side of the roll; and a guide portion which extends in the width direction at a backside, in a horizontal direction, of the roll supported by the first roll support portion and the second roll support portion and movably supports the second roll support portion in the width direction, wherein the second roll support portion has a roller portion in contact with the guide portion and being rotatable around a rotation axis extending in a direction orthogonal to the width direction, and wherein the roller portion includes a first roller in contact with the guide portion from an upper side in a vertical direction, a second roller in contact with the guide portion from a rear side with respect to the front side, and a third roller in contact with the guide portion below the second roller from a fore side with respect to the rear side.
2. The recording device according to claim 1, wherein the recording device further includes: a third roll support portion which rotatably supports another roll on the one end side of the other roll; and a fourth roll support portion which is provided to face the third roll support portion and rotatably supports the other roll on the other end side of the other roll, and wherein the sheet for which the recording operation was performed is taken up by the other roll installed between the third roll support portion and the fourth roll support portion.
3. The recording device according to claim 2, wherein a rotation center of the roll supported by the first roll support portion and the second roll support portion matches a rotation center of the other roll supported by the third roll support portion and the fourth roll support portion in height.
4. The recording device according to claim 1, wherein below a first space in which the roll between the first roll support portion and the second roll support portion is installed, a second space into

which a cart which transports the roll can enter is formed.
5. The recording device according to claim 4, wherein the guide portion is located on a downstream side in an entry direction of the cart with respect to the first space.
6. The recording device according to claim 5, further comprising: a guide support portion which supports the guide portion; and a position adjustment mechanism capable of adjusting a position in the entry direction and a vertical direction of at least one of the first roll support portion and the second roll support portion with respect to the guide support portion.
7. The recording device according to claim 6, wherein the position adjustment mechanism includes a connection member which connects the guide portion and the guide support portion and can adjust the position in the entry direction and the vertical direction of at least either one of the first roll support portion and the second roll support portion with respect to the guide support portion.
8. The recording device according to claim 6, wherein the position adjustment mechanism includes a slide member which extends in the width direction, is provided on the guide portion, and can adjust the position in the entry direction and the vertical direction of at least either one of the first roll support portion and the second roll support portion with respect to the guide portion by being brought into contact with the second roll support portion.
9. The recording device according to claim 8, wherein the slide member includes a first slider provided on an upper end of the guide portion, a second slider provided on a downstream side in the entry direction with respect to the guide portion, and a third slider provided below the second slider and on an upstream side in the entry direction with respect to the guide portion.
10. The recording device according to claim 1, further comprising: a first stand configured to support the recording portion at the first roll support portion side; and a second stand configured to support the recording portion at the second roll support portion side, wherein the guide portion is attached to the first stand and the second stand.
11. The recording device according to claim 1, further comprising: a lock mechanism configured to fix the second roll support portion to the guide portion.
12. The recording device according to claim 1, further comprising: a roll drive motor configured to rotate the roll at the first roll support portion side.
13. The recording device according to claim 1, wherein the recording portion is an inkjet type printhead.